Harvesting under a Strong Allee Effect: Collapse Thresholds, Hysteresis and Comparison of Constant-Yield vs Proportional-Effort Harvesting, with Physics-Informed Neural Network (PINN) Forward and Inverse Analysis

RaghavSinghal15, AshChan07, devam2006
Prem-Mehta03, rohanagrawal28
Author names await final confirmation

Working draft, 28 October 2026

Abstract

We compare fixed-catch and proportional-effort harvesting in a strong-Allee population model. The implemented classical analysis derives harvesting thresholds and initial-state basins, calculates finite A extinction times, and checks numerical references against independent methods. The example $r = 0.7$, $K = 100$, $A = 20$ gives $H_c = 36.75966303$ population/time and $h_c = 0.56$ inverse time. These have different units and do not directly rank policy risk. **SCAFFOLD_ONLY:** Full comparison/history and PINN forward/inverse evidence remain to be integrated; this is not the final report.

Working draft and scope

This is an incomplete report for group review. The WP03 classical/numerical analysis is implemented; the remaining sections retain explicit gaps until their owners supply reviewed work. No forward or inverse PINN performance is claimed in this draft. The numerical examples are synthetic.

Introduction and objectives

We study harvesting in a population with a minimum viable density, represented by a strong Allee effect. The aim is to derive collapse thresholds, distinguish the role of harvesting from that of initial population, and test numerical and PINN methods using reproducible evidence. The project compares constant catch with proportional effort, examines harvesting history and recovery barriers, and evaluates forward/inverse PINNs once the corresponding experiments are available.

Assume one deterministic, well-mixed population with constant $r > 0$, $0 < A < K$, nonnegative harvesting, and no age structure or delays. Write

$$g(N) = rN(1 - N/K)(N/A - 1), \qquad N' = g(N) - H \text{ (A)}, \qquad N' = g(N) - hN \text{ (B)}.$$

Here H has population/time units and h inverse-time units. For forward analysis use $x = N/K$, $\tau = rt$, $a = A/K$, $q = H/(rK)$ and $e = h/r$. Unknown-parameter inverse analysis must instead respect fixed external scaling. The equations are the group's approved model; their illustrative parameters are not calibrated to a measured species.

1 Foundations and Model A

SCAFFOLD_ONLY: Units and assumptions; logistic baseline; nondimensionalisation; A equilibria and stability, phase lines and branch plot.

Replace this template with derivations, algorithms, labelled figures/tables, interpretation, limitations and verified references from WP01.

2 Model B, comparison and harvesting history

SCAFFOLD_ONLY: B roots/stability; fair A/B comparisons; up/down ramp with state carry-over; explicit restocking; no invented automatic recovery.

Replace this template with derivations, algorithms, labelled figures/tables, interpretation, limitations and verified references from WP02.

3 Collapse thresholds, extinction and basins

All examples below are synthetic, under a deterministic, well-mixed population with constant parameters, no age structure or delays, $r > 0$, $0 < A < K$ and nonnegative harvest. Let $g(N) = rN(1 - N/K)(N/A - 1)$. Setting $x = N/K$, $\tau = rt$, $a = A/K$, $q = H/(rK)$, $e = h/r$ gives

$$x' = G(x) - q \quad \text{(A)}, \qquad x' = G(x) - ex \quad \text{(B)}, \qquad G(x) = -x^3/a + (1 + a)x^2/a - x.$$

Here prime means differentiation with respect to dimensionless time τ.

3.1 Critical controls from calculus

Differentiation gives $G'(x) = [-3x^2 + 2(1 + a)x - a]/a$. Write $d = \sqrt{1 - a + a^2}$. The minus stationary point $(1 + a - d)/3$ lies below a; the plus stationary point $x_* = (1 + a + d)/3$ lies in $(a, 1)$ and has $G''(x_*) = -2d/a < 0$. Consequently

$$q_c = G(x_*), \quad N_* = \frac{K + A + \sqrt{K^2 - KA + A^2}}{3}, \quad H_c = rKq_c.$$

For B, a positive equilibrium satisfies $e = P(x) = (1 - x)(x/a - 1)$. Since $P'(x) = (1 + a - 2x)/a$ and $P'' = -2/a$, its maximum is at $(1 + a)/2$, giving

$$e_c = \frac{(1 - a)^2}{4a}, \qquad h_c = \frac{r(K - A)^2}{4AK}.$$

The class logistic maximum is instead $1/4$ at $x = 1/2$, so $H_{c,\text{logistic}} = rK/4$. Equal r, K do not calibrate these different production functions. Also H has population/time units while h has inverse-time units; their numerical values cannot be compared directly.

For $r = 0.7$, $K = 100$, $A = 20$, the example gives $N_* = 70.55050463$, $H_c = 36.75966303$, $h_c = 0.56$ and $H_{c,\text{logistic}} = 17.5$ in declared model units. Bounded numerical maximisation checks both critical controls for six a values from 0.05 to 0.95; see `wp03_thresholds.csv`.

3.2 Basins and exact critical cases

For $0 < q < q_c$, A has two positive roots $x_- < x_+$ in $(a, 1)$. The signs of $G - q$ are negative, positive and negative on the three intervals: the lower root is unstable and the upper root stable. The negative cubic root is excluded from the population domain. For B the positive roots are

$$x_\pm = \frac{1 + a \pm \sqrt{(1 - a)^2 - 4ae}}{2}, \qquad f_B'(0) = -(1 + e) < 0.$$

At either positive fold the roots coalesce. Although the derivative is zero, the vector field is negative on both sides locally: the double root is semistable, attracting from above and departing from below.

Control	A: constant yield	B: effort
Zero	Below a approaches zero; above a tends to 1	Same positive-state basins; zero stable
Below fold	Below x_- hits zero; above x_- tends to x_+	Below x_- approaches zero; above tends to x_+
At fold	Below x_* hits zero; above approaches x_*	Below double root approaches zero; above approaches double root
Above fold	Every finite positive start hits zero	Every positive start approaches zero

Exact initial equilibria remain there in every row. A with $q > 0$ has $f_A(0) = -q$: zero is not an equilibrium; initial zero records immediate boundary termination. At $q = 0$ and for B, zero is an equilibrium.

The classifier uses sign arguments, not a finite solver horizon. Exact floating matches to returned roots are recognised. Non-exact starts within $2\,10^{-12}$ dimensionless population units of an unstable or semistable boundary are labelled unresolved. Controls are not merged by a fold tolerance: adjacent representable controls below and above the computed cliff remain distinct. These are finite-precision conventions, not measurement uncertainty. WP01/WP02 retain ownership of their full equilibrium/stability presentations.

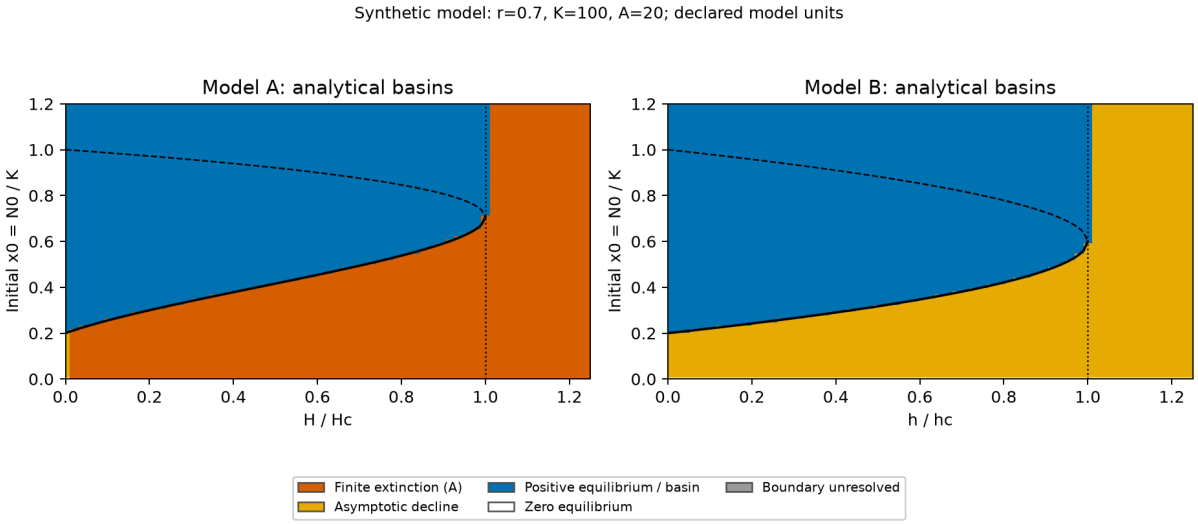


Figure 1: Synthetic analytical basin grids. Solid black curves are basin boundaries, dashed curves upper positive equilibria, dotted vertical lines folds. A decline under positive yield reaches zero in finite time; B decline is asymptotic. The exact equilibrium lines and thin zero-harvest column also require the classification table; the colour grid has finite resolution.

3.3 Extinction quadrature and numerical checks

On a reaching-zero A path, separation gives

$$\tau_{\text{ext}} = \int_0^{x_0} \frac{du}{q - G(u)}, \qquad t_{\text{ext}} = \tau_{\text{ext}}/r = \int_0^{N_0} \frac{dN}{H - g(N)}.$$

Classify before integrating. For a subcritical reaching-zero start, the denominator is continuous and strictly positive on $[0, x_0]$, giving finite time. For $q > q_c$, $G(x) \le q_c$ for all $x \ge 0$ additionally

gives $\tau_{\text{ext}} \le x_0/(q - q_c)$. An intervening equilibrium is a pole and makes this extinction formula inapplicable. With $q = 0$, smoothness and uniqueness exclude a positive state arriving at the zero equilibrium in finite time. The same argument applies to B; no B cutoff time is claimed as exact extinction.

Near a subcritical basin boundary, factor the cubic as $(x - x_n)(x_- - x)(x_+ - x)/a$, where $x_n < 0$. For the difficult upper portion of the integral, the substitution $u = \log[x_-/(x_- - x)]$ removes the vanishing factor and avoids cancellation. At the computed fold, put $s = x_* - x_0 > 0$ and $d = \sqrt{1 - a + a^2}$. Partial fractions give an elementary evaluation of the same integral:

$$\tau_{\text{ext}} = \frac{a x_0}{d s x_*} + \frac{a}{d^2} \log\left(1 + \frac{d x_0}{s(d - x_*)}\right).$$

This avoids missed endpoint contributions for starts extremely close below the fold. Its method is recorded as `fold_closed_form`, with no invented quadrature-error estimate. Root rounding remains a separate source of uncertainty near a basin boundary.

SciPy RK45 uses the Dormand–Prince 5(4) pair: shared stage slopes produce two estimates, their difference controls local error, and the fifth-order value advances the state[3]. The wrapper stops at the first downward zero event for positive-yield A and includes the event endpoint. No post-event zero tail is passed to the original ODE. Output times do not prescribe internal adaptive steps. Default reference tolerances are 10^{-9} relative, 10^{-11} absolute, maximum dimensionless step 0.1.

The 427 initial-state/control evaluations retain explicit outcome labels; a blank time means no finite extinction or unresolved classification, not zero or a finite-horizon estimate. Six representative finite paths, including the fold from below, are checked by the separated integral (quadrature or the fold expression above), RK45 at relative tolerances 10^{-8} and 10^{-10}, and DOP853 at 10^{-12}. The latter integrates the independently expanded polynomial $-q - x + (1 + a)x^2/a - x^3/a$. Acceptance is $|\Delta\tau| \le 2\,10^{-7} + 5\,10^{-7}\tau_{\text{ext}}$; the largest observed discrepancy is below $4.6\,10^{-8}$. QUADPACK's error output is an estimate, not a rigorous bound[2]. Tests additionally cover several a values, physical units, exact logistic solutions and deliberate corruption of the production RHS.

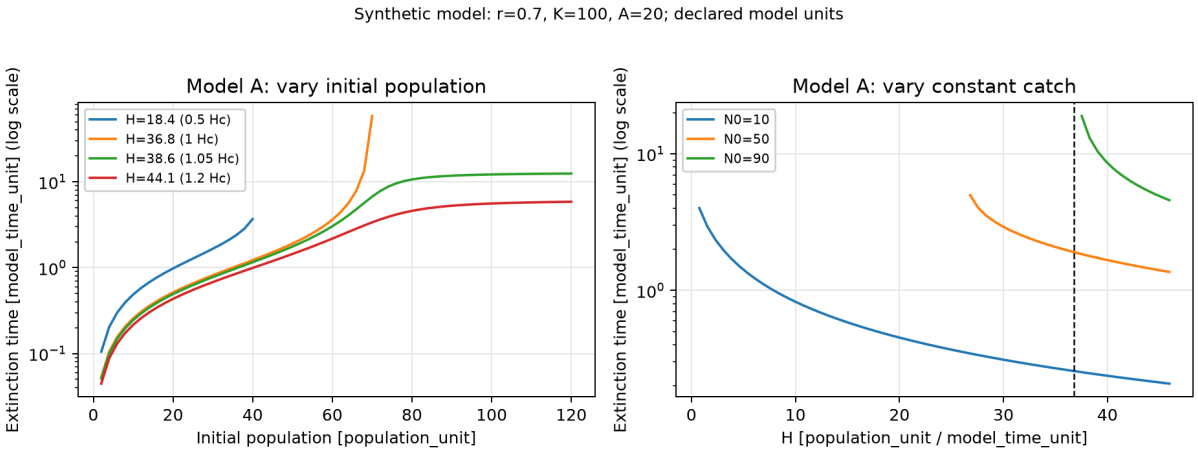


Figure 2: A extinction time versus initial state and catch in declared model units; vertical axes are logarithmic. Gaps mean no finite extinction, equilibrium or unresolved boundary; initial zero has time zero and is omitted. The dashed line on the right is H_c. Times grow near a basin boundary and decrease with catch along a fixed reaching-zero initial state.

3.4 Bottleneck above the fold

Set $q = q_c + \delta$, $\delta > 0$, and $y = x - x_*$. The cubic expansion is exact:

$$q - G(x) = \delta + \beta y^2 + y^3/a, \qquad \beta = d/a > 0.$$

For a fixed start above x_* and a terminal state below it, the leading passage contribution follows from

$$\int \frac{dy}{\delta + \beta y^2} = \frac{\arctan(y\sqrt{\beta/\delta})}{\sqrt{\beta\delta}}, \qquad \tau_{\text{ext}} \sim \frac{\pi}{\sqrt{\beta\delta}} \quad (\delta \downarrow 0).$$

This is passage after the positive equilibria disappear, distinct from local relaxation toward a stable equilibrium. A fixed start below x_* does not cross this bottleneck and need not have this leading divergence. At the fold itself, a start above approaches algebraically: $y' \simeq -\beta y^2$ gives $y \simeq y_0/(1 + \beta y_0 \tau)$. A start below still reaches zero.

The substitution $y = \sqrt{\delta/\beta}\tan\theta$ resolves the full near-fold integral, with transformed integrand $[\sqrt{\beta\delta}(1 + (y/d)\sin^2\theta)]^{-1}$. At $x_0 = 1.1$, the example's measured $\tau_{\text{ext}}\sqrt{\delta}$ rises from 1.36535 at $\delta/q_c = 0.1$ to 1.46542 at 10^{-4}, approaching the predicted 1.46756.

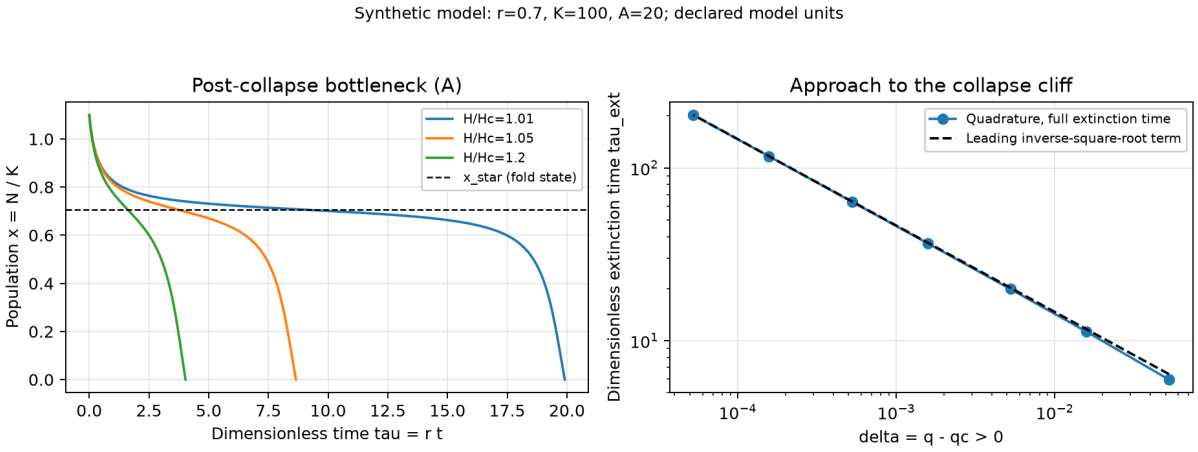


Figure 3: Synthetic supercritical A trajectories and bottleneck divergence. Both axes on the right are dimensionless and logarithmic. The dashed small-δ expression is a leading asymptotic, not an exact finite-δ fit. All trajectories terminate at their zero event.

3.5 Optional sensitivity and sustainable yield

Holding the other dimensional parameters fixed, $g_N(N_*) = 0$ removes the derivative of N_* from dH_c:

$$H_{c,r} = H_c/r, \quad H_{c,K} = rN_*^2(N_*/A - 1)/K^2, \quad H_{c,A} = -rN_*^2(1 - N_*/K)/A^2.$$

Since $h_c = \frac{r}{4}(K/A - 2 + A/K)$,

$$h_{c,r} = h_c/r, \quad h_{c,K} = \tfrac{r}{4}(1/A - A/K^2), \quad h_{c,A} = \tfrac{r}{4}(-K/A^2 + 1/K).$$

Elasticity is $(p/C)\partial C/\partial p$. For the example, the (r, K, A) elasticities are $(1, 2.39564, -1.39564)$ for H_c and $(1, 1.5, -1.5)$ for h_c. Central differences check the partials; population-scale homogeneity makes the last two elasticities sum to 1 and 0 respectively. An a sweep at fixed r, K is a different experiment.

For A the stable positive branch has yield approaching the supremum H_c, at the semistable fold with zero control margin. For B, $Y(h) = hN_+(h) = g(N_+(h))$. Since $N_* > (K + A)/2$, the production-maximising state is on its stable upper branch. Hence

$$h_{\text{MSY}} = g(N_*)/N_* < h_c, \qquad Y_{\text{MSY}} = H_c.$$

The example gives $h_{\mathrm{MSY}} = 0.5210404$ and effort margin $1 - h_{\mathrm{MSY}}/h_c = 0.0695707$. Independent bounded maximisation checks this interior optimum[1]. The initial population must still lie above the unstable root.

At $N_0 = 90$, illustrative policies A at $H = 0.8H_c$ and B at its MSY give population margins $N_0 - N_- = 36.1954$ and 40.5505, with control margins 20% and 6.957%. Their initial realised catches differ; these separate examples do not establish a universal risk ranking.

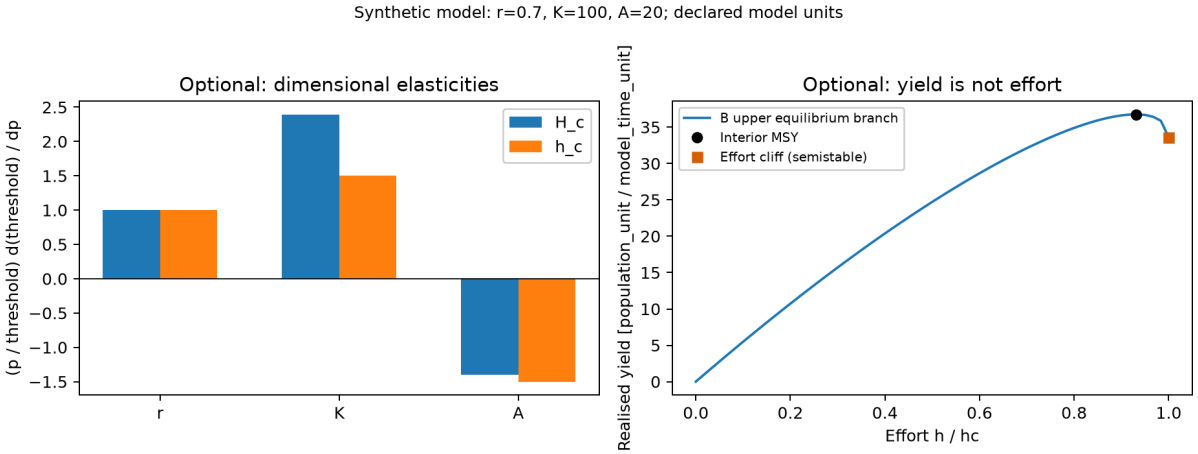


Figure 4: Optional dimensional elasticities and B upper-branch yield. The interior MSY is distinct from the semistable effort cliff. Yield uses declared population/time units. Each elasticity holds the other dimensional parameters fixed.

3.6 Reproduction, handoff and limitations

The versioned WP03 experiment regenerates the same tables and figures used here and in the notebook; manifests record config, versions and source/file hashes. The early reference handoff remains six cases and 606 clean synthetic points, with known conditions separate from unknown evaluation truth. WP05 owns observation selection, noise, frozen splits, inverse fitting, baseline and held-out evaluation; WP04 supplies the neural engine. No field-data or species-specific validation is claimed.

The submission audit lists unfinished group sections and missing final artifacts/metadata. Group packaging waits for the other work packages, reviews and reproduction; this section does not certify their completion. AI assistance: OpenAI ChatGPT/Codex for implementation, exposition, testing and documentation, recorded in the project acknowledgement log. Devam's detailed learning review remains pending.

4 Forward PINNs

SCAFFOLD_ONLY: Differentiable network and residual; initial condition/physics loss; optimiser/collocation algorithm; both models and independent solver errors/runtime.

Replace this template with derivations, algorithms, labelled figures/tables, interpretation, limitations and verified references from WP04.

5 Inverse PINNs and validation

SCAFFOLD_ONLY: Identifiable staged unknowns; noisy observations; ODE least-squares baseline; held-out prediction; near-collapse failures; AI/data acknowledgement.

Replace this template with derivations, algorithms, labelled figures/tables, interpretation, limitations and verified references from WP05.

6 Conclusions and limitations

The current classical results show that harvesting thresholds alone do not determine persistence: the initial population must also exceed its basin barrier. Positive fixed catch permits finite-time extinction, whereas B extinction paths from positive starts approach zero asymptotically. Near the fold, supercritical trajectories can spend a long time in a bottleneck before collapse. These statements concern the assumed deterministic model and synthetic examples. **SCAFFOLD_ONLY:** Integrate the fair policy/history comparison, quantitative PINN evidence and failure cases before finalising the project's conclusions.

7 Contributions and acknowledgements

The task allocation is WP01: `RaghavSinghal15`; WP02: `AshChan07`; WP03: `devam2006`; WP04: `Prem-Mehta03`; WP05: `rohanagrawal28`. OpenAI ChatGPT/Codex assisted with repository preparation, WP03 implementation, exposition, checks and this working assembly, as recorded in the AI-use log. **SCAFFOLD_ONLY:** Confirm actual author names and each member's contribution, verification and AI/source disclosures before submission.

References

[1] SciPy developers. Bounded scalar minimisation documentation. `https://docs.scipy.org/doc/scipy/reference/optimize.minimize_scalar-bounded.html`. Accessed 28 October 2026.

[2] SciPy developers. quad documentation. `https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.quad.html`. Accessed 28 October 2026.

[3] SciPy developers. solve_ivp documentation. `https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve_ivp.html`. Accessed 19 October 2026.